

PROBLEM SET 2: EXTENSIONS OF THE RAMSEY MODEL

PROBLEM I – THE AK MODEL WITH AN EXTERNALITY

Consider the following version of the Ramsey model:

$$U_0 = \int_0^\infty e^{-\rho t} \ln C_t dt, \quad (1)$$

$$Y_t = A_t K_t^\alpha L_t^{1-\alpha}, \quad (2)$$

$$\dot{K}_t = Y_t - C_t, \quad (3)$$

$$A_t = AK_t^{1-\alpha}, \quad (4)$$

$$L_t = 1. \quad (5)$$

In this set-up, the level of technology A_t is *endogenous* and depends on the aggregate capital stock. Private agents do not take into account equation (4) when making their investment decision. Note also that the exponent on K in (4) is such that there are constant returns to capital in the aggregate, so that the aggregate production function writes

$$Y_t = AK_t.$$

- 1 – Comment equation (4). Do the competitive equilibrium and the Pareto optimum coincide in this economy?
- 2 – Set up the planning problem and solve for the Pareto optimal allocation. What is the optimal growth rate of the economy?
- 3 – We now consider the competitive equilibrium. Let r_t and w_t denote the real rental rate of capital and the real wage. Solve the problem of the representative firm to obtain the equilibrium values of r_t and w_t .
- 4 – Letting a_t denote the (per-capita) asset position of the representative consumer, write the flow budget constraint and the consumer's problem. Show that the Euler equation is

$$\frac{\dot{C}_t}{C_t} = \alpha A - \rho.$$

- 5 – Letting $c_t = C_t/K_t$, show that c_t follows a linear differential equation which is unstable in the usual sense.
- 6 – What is the value of c_t along a balanced growth path? What is the growth rate of the economy in the competitive equilibrium?
- 7 – Show that the economy is on a balanced growth path from date 0 on, regardless of the value of K_0 .
- 8 – Is growth too high or too low in the competitive equilibrium? Explain.

PROBLEM II – GROWTH WITH INVESTMENT-SPECIFIC TECHNOLOGICAL CHANGE

This problem considers an alternative conception of technology, developed for instance in Greenwood, Hercowitz, and Krusell (American Economic Review, 1997). The representative firm produces according to

$$Y_t = A_t K_t^\alpha L_t^{1-\alpha},$$

where A_t is an exogenous neutral technology shifter evolving according to

$$\frac{\dot{A}_t}{A_t} = g_A \geq 0.$$

The capital accumulation equation writes

$$\dot{K}_t = I_t - \delta K_t,$$

and the economy's resource constraint is

$$Y_t = C_t + \frac{I_t}{X_t},$$

where X_t is an exogenous process evolving according to

$$\frac{\dot{X}_t}{X_t} = g_X \geq 0.$$

The representative consumer's utility is

$$U_0 = \int_0^\infty e^{-\rho t} \ln C_t dt,$$

and population is constant and normalized to $L = 1$.

- 1 – Comment the resource constraint. How would you interpret X_t ?
- 2 – Set up the planning problem and obtain the Euler equation for consumption.
- 3 – What are the growth rates of Y_t , C_t , I_t , and K_t along a balanced growth path?
- 4 – Without solving for the competitive equilibrium, argue that the relative price of investment P_t^I/P_t^C is equal to $1/X_t$ in equilibrium. Gordon (*The Measurement of Durable Goods Prices*, University of Chicago Press, 1990) argues that P^I/P^C has been falling throughout the postwar period in the US. What does this fact imply for the value of g_X and for the sources of economic growth?
- 5 – Show that K/Y is not constant along the balanced growth path. Does that violate one of Kaldor's stylized fact? (Hint: What is the value of the capital stock in terms of the final good?)

PROBLEM III – THE AUGMENTED SOLOW MODEL WITH ONLY ACCUMULABLE FACTORS (*easy, do it yourself*)

Consider the following model:

$$Y = AK^\alpha H^{1-\alpha},$$

where Y = output, K = capital, H = human capital
and:

$$\begin{aligned}\dot{K} &= s_K Y, \\ \dot{H} &= s_H Y.\end{aligned}$$

- 1 – Show that if (K_t, H_t) is an equilibrium trajectory for initial conditions (K_0, H_0) , then for any $\lambda > 0$ we have that $(\lambda K_t, \lambda H_t)$ is an equilibrium trajectory for initial conditions $(\lambda K_0, \lambda H_0)$.
- 2 – Show that the equilibrium ratio H/K converges to a unique long-term level. Compute this level.
- 3 – Compute the long-term growth rate of this economy.
- 4 – Under which conditions will the economy be on a balanced growth path from $t = 0$ on?